Carbon Footprint Prediction using Machine Learning

Using ML to measure and reduce individual environmental impact

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Problem & Objective

Most people are unaware of their daily carbon footprint. Manual calculation is complex and time-consuming. This project uses machine learning to solve that problem.

Predict

Estimate carbon footprint value from user behavior data

Classify

Categorize impact as **High / Medium / Low**

Identify

Find the key factors that affect the environment most

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Data & Preprocessing

Dataset

Personal Carbon Footprint Behavior — 1,400 records with lifestyle and consumption data

Preprocessing Steps

- Removed unnecessary columns: `user_id` and `carbon_footprint_kg` (for classification task)
- Checked for missing values — **none found**
- Applied **one-hot encoding** to convert categorical data into numerical format
- Split data: **80% training** and **20% testing**

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Model & Results

Two models were trained — a **Random Forest Classifier** for impact category and a **Linear Regression** model for footprint value prediction.

Random Forest Classifier

Accuracy ≈ 93%

Correctly classifies High / Medium / Low impact

Cross Validation ≈ 94%

Consistent performance across data folds

Linear Regression

MAE ≈ 0.41

Low average prediction error

RMSE ≈ 0.55

Strong precision in value estimates

R^2 ≈ 0.95

Explains 95% of variance in data

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Conclusion & Future Work

✅ What We Achieved

- Model successfully predicts carbon footprint with high accuracy
- Helps users understand their environmental impact clearly
- Strong performance across both classification and regression tasks

🚀 Future Work

- Build a mobile or web application for public use
- Enable real-time carbon tracking from daily activities
- Suggest personalized eco-friendly habits to users

> **Thank You**

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